CS5275 – The Algorithm Designer's Toolkit (S2 AY2025/26)

Lecture 13:

Boolean functions – Arrow's impossibility theorem

Nicolas de Condorcet

Condorcet paradox

- **Majority rule:**
  - If most people prefer a over b, then a wins.
- But what happens when there are more than 2 candidates?

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- But what happens when there are more than 2 candidates?

Setup: 3 voters rank 3 candidates: a, b, c
- Voter 1: $a > b > c$
- Voter 2: $b > c > a$
- Voter 3: $c > a > b$

Majority rule →

Condorcet cycle:
- a vs. b: a wins.
- b vs. c: b wins.
- c vs. a: c wins.

Nicolas de Condorcet

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Majority rule

Condorcet cycle:
- a vs. b: a wins.
- b vs. c: b wins.
- c vs. a: c wins.

- **Question:** Can we avoid Condorcet cycles by using other rules?

Impossibility

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- Arrow proved a surprising result.
  - If there are more than 2 candidates, no voting rule can satisfy all of the following:

Pareto efficiency:
- If everyone prefers a over b, the group should prefer a over b.

$\longleftrightarrow$

Unanimous:
$f(1, \ldots, 1) = 1$ and $f(-1, \ldots, -1) = -1$

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Independence of irrelevant alternatives:
- The ranking between a and b should depend only on how people rank a vs. b.

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Take-away messages:
- Designing a perfectly fair voting system is impossible.
- Trade-offs between fairness principles are unavoidable.

Impossibility

Arrow later received the **Nobel Memorial Prize in Economic Sciences**, partly for this work.
https://www.nobelprize.org/prizes/economic-sciences/1972/arrow/facts/

This insight helped create the field of **social choice theory**.

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Take-away messages:
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Setting

Voters' preferences

	#1	#2	#3	$\cdots$		**Outcome**
a (1) vs. b (−1)	1	1	−1	$\cdots$	$= x$	$f(x)$
b (1) vs. c (−1)	1	−1	1	$\cdots$	$= y$	$g(y)$
c (1) vs. a (−1)	−1	−1	1	$\cdots$	$= z$	$h(z)$

Independence of irrelevant alternatives:

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For simplicity, in the lecture we assume $f = g = h$.

We leave the extension to allow different functions as an exercise.

Independence of irrelevant alternatives:

- The ranking between a and b should depend only on how people rank a vs. b.

Setting

$$\mathrm{NAE}_3(s_1, s_2, s_3) = \begin{cases} 0 & s_1 = s_2 = s_3 \\ 1 & \text{otherwise} \end{cases}$$

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There are $3! = 6$ possible rankings of 3 candidates, corresponding to 6 possible values of (x_i, y_i, z_i):

$$\underbrace{(1,1,-1), \quad (1,-1,-1), \quad (-1,1,-1), \quad (-1,1,1), \quad (1,-1,1), \quad (-1-1,1)}$$

These are precisely the triples satisfying the **not-all-equal** predicate NAE_3.

Setting

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Condorcet paradox:

- $f = \mathrm{Maj}_n$ does not work.

A candidate is a **Condorcet winner** if it wins all of the pairwise elections in which it participates.

$$\exists \text{ Condorcet winner} \longleftrightarrow \mathrm{NAE}_3\big(f(x), f(y), f(z)\big) = 1$$

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Arrow's theorem: Suppose $f : \{-1,1\}^n \to \{-1,1\}$ is a unanimous voting rule.

- If there is always a Condorcet winner, then f must be a dictatorship.

Proof idea:

- Analyze $\Pr[\exists \text{ Condorcet winner}] = \mathbb{E}\big[\mathrm{NAE}_3\big(f(\boldsymbol{x}), f(\boldsymbol{y}), f(\boldsymbol{z})\big)\big]$

Analysis

$$\mathrm{NAE}_3(s_1, s_2, s_3) = \begin{cases} 0 & s_1 = s_2 = s_3 \\ 1 & \text{otherwise} \end{cases}$$

Voters' preferences

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- $f = \mathrm{Maj}_n$ does not work.

Each voter chooses its ranking uniformly at random, over the 6 possible rankings of 3 candidates:

$$(\boldsymbol{x}_i, \boldsymbol{y}_i, \boldsymbol{z}_i) \sim \{(1,1,-1), (1,-1,-1), (-1,1,-1), (-1,1,1), (1,-1,1), (-1-1,1)\}$$

Analysis

$$\mathrm{NAE}_3(s_1,s_2,s_3) = \begin{cases} 0 & s_1 = s_2 = s_3 \\ 1 & \text{otherwise} \end{cases}$$

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$(\boldsymbol{x},\boldsymbol{y}),(\boldsymbol{y},\boldsymbol{z}),(\boldsymbol{z},\boldsymbol{x})$ are $(-1/3)$-correlated.

$(\boldsymbol{x},\boldsymbol{y})$ is a ρ-correlated pair of random strings:

- $\boldsymbol{x} \sim \{-1,1\}^n$
- $\boldsymbol{y} \sim \{-1,1\}^n$
- $\mathbb{E}[\boldsymbol{x}_i \boldsymbol{y}_i] = \rho$

Analysis

$$\mathrm{NAE}_3(s_1, s_2, s_3) = \begin{cases} 0 & s_1 = s_2 = s_3 \\ 1 & \text{otherwise} \end{cases}$$

$\Pr[\exists \text{ Condorcet winner}]$
$= \mathbb{E}\left[\mathrm{NAE}_3\big(f(\boldsymbol{x}), f(\boldsymbol{y}), f(\boldsymbol{z})\big)\right]$
$= \frac{3}{4} - \frac{1}{4}\left(\mathbb{E}[f(\boldsymbol{x})f(\boldsymbol{y})] + \mathbb{E}[f(\boldsymbol{y})f(\boldsymbol{z})] + \mathbb{E}[f(z)f(\boldsymbol{x})]\right)$
$= \frac{3}{4}\left(1 - \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ (-1/3)\text{-correlated}}[f(\boldsymbol{x})f(\boldsymbol{y})]\right)$

$$\mathrm{NAE}_3(s_1, s_2, s_3) = \frac{3}{4} - \frac{1}{4}(s_1 s_2 + s_1 s_3 + s_2 s_3)$$

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Lemma: (when $-1 < \rho < 0$)

$\boxed{\mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{-correlated}}[f(\boldsymbol{x})f(\boldsymbol{y})] = \rho}$

$\Updownarrow$

$\boxed{f \text{ is a dictator or its negation}}$

$\Pr[\exists \text{ Condorcet winner}] = 1$

$\Updownarrow$

f is a $\underbrace{\textbf{dictator} \text{ or its negation.}}$

Impossible because f is **unanimous.**

$\uparrow$

Pareto efficiency

$(\boldsymbol{x}, \boldsymbol{y})$ is a ρ-correlated pair of random strings:

- $\boldsymbol{x} \sim \{-1,1\}^n$
- $\boldsymbol{y} \sim \{-1,1\}^n$
- $\mathbb{E}[\boldsymbol{x}_i \boldsymbol{y}_i] = \rho$

Noise stability

Noise stability of $f : \{-1,1\}^n \to \mathbb{R}$ at ρ is

$$\mathrm{Stab}_\rho(f) = \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{-correlated}}[f(\boldsymbol{x})f(\boldsymbol{y})].$$

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- $\boldsymbol{x} \sim \{-1,1\}^n$
- $\boldsymbol{y} \sim \{-1,1\}^n$
- $\mathbb{E}[\boldsymbol{x}_i \boldsymbol{y}_i] = \rho$

$\Pr[\boldsymbol{y}_i = \boldsymbol{x}_i] = \frac{1}{2}(1+\rho)$ $\quad$ $\Pr[\boldsymbol{y}_i = -\boldsymbol{x}_i] = \frac{1}{2}(1-\rho)$ $\quad \longleftrightarrow$

Noise stability

$\mathrm{Stab}_\rho(\chi_S) = \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{-correlated}}[\chi_S(\boldsymbol{x})\chi_S(\boldsymbol{y})] = \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{-correlated}}[\prod_{i\in S} \boldsymbol{x}_i\boldsymbol{y}_i] = \prod_{i\in S} \mathbb{E}[\boldsymbol{x}_i\boldsymbol{y}_i] = \rho^{|S|}$

Noise stability of $f : \{-1,1\}^n \to \mathbb{R}$ at ρ is

$\mathrm{Stab}_\rho(f) = \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{-correlated}}[f(\boldsymbol{x})f(\boldsymbol{y})]$.

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Noise stability

$$\mathrm{Stab}_\rho(\chi_S) = \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{–correlated}}[\chi_S(\boldsymbol{x})\chi_S(\boldsymbol{y})] = \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{–correlated}}[\prod_{i\in S}\boldsymbol{x}_i\boldsymbol{y}_i] = \prod_{i\in S}\mathbb{E}[\boldsymbol{x}_i\boldsymbol{y}_i] = \rho^{|S|}$$

$$\begin{aligned}
&\mathrm{Stab}_\rho(f)\\
&= \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{–correlated}}[f(\boldsymbol{x})f(\boldsymbol{y})]\\
&= \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{–correlated}}\left[\sum_{S,T\subseteq[n]}\hat{f}(S)\hat{f}(T)\chi_S(\boldsymbol{x})\chi_T(\boldsymbol{y})\right]\\
&= \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{–correlated}}\left[\sum_{S\subseteq[n]}\hat{f}(S)^2\chi_S(\boldsymbol{x})\chi_S(\boldsymbol{y})\right]\\
&= \sum_{S\subseteq[n]}\hat{f}(S)^2\rho^{|S|}
\end{aligned}$$

Noise stability of $f : \{-1,1\}^n \to \mathbb{R}$ at ρ is

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Noise stability

$\mathrm{Stab}_\rho(f)$

$= \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{-correlated}}[f(\boldsymbol{x})f(\boldsymbol{y})]$

$= \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{-correlated}}\left[\sum_{S,T\subseteq[n]} \hat{f}(S)\hat{f}(T)\chi_S(\boldsymbol{x})\chi_T(\boldsymbol{y})\right]$

$= \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{-correlated}}\left[\sum_{S\subseteq[n]} \hat{f}(S)^2\chi_S(\boldsymbol{x})\chi_S(\boldsymbol{y})\right]$

$= \sum_{S\subseteq[n]} \hat{f}(S)^2\rho^{|S|} = \sum_k \rho^k \cdot \mathrm{W}^k[f]$

$$\boxed{\mathrm{W}^k[f] = \sum_{S\subseteq[n]\,:\,|S|=k} \hat{f}(S)^2}$$

Noise stability

$$\sum_k \mathrm{W}^k[f] = 1$$

Equality holds iff $\mathrm{W}^1[f] = 1$

$$\sum_k \rho^k \cdot \mathrm{W}^k[f] \geq \rho \cdot \sum_k \mathrm{W}^k[f] = \rho$$

ρ^k is minimized when $k = 1$.

$$\begin{aligned}
&\mathrm{Stab}_\rho(f) \\
&= \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{-correlated}}[f(\boldsymbol{x})f(\boldsymbol{y})] \\
&= \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{-correlated}}\left[\sum_{S,T\subseteq[n]} \hat{f}(S)\hat{f}(T)\chi_S(\boldsymbol{x})\chi_T(\boldsymbol{y})\right] \\
&= \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{-correlated}}\left[\sum_{S\subseteq[n]} \hat{f}(S)^2\chi_S(\boldsymbol{x})\chi_S(\boldsymbol{y})\right] \\
&= \sum_{S\subseteq[n]} \hat{f}(S)^2\rho^{|S|} = \sum_k \rho^k \cdot \mathrm{W}^k[f]
\end{aligned}$$

$$\boxed{\mathrm{Stab}_\rho(f) = \rho} \Leftrightarrow \boxed{\mathrm{W}^1[f] = 1}$$

for $f : \{-1,1\}^n \to \{-1,1\}$, $-1 < \rho < 0$

$$\boxed{\mathrm{W}^k[f] = \sum_{S\subseteq[n]\,:\,|S|=k} \hat{f}(S)^2}$$

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&= \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\ \rho\text{-correlated}}\left[\sum_{S\subseteq[n]} \hat{f}(S)^2\chi_S(\boldsymbol{x})\chi_S(\boldsymbol{y})\right] \\
&= \sum_{S\subseteq[n]} \hat{f}(S)^2\rho^{|S|} = \sum_k \rho^k \cdot \mathrm{W}^k[f]
\end{aligned}$$

$$\boxed{\mathrm{W}^k[f] = \sum_{S\subseteq[n]\,:\,|S|=k} \hat{f}(S)^2}$$

$$\boxed{\mathrm{Stab}_\rho(f) = \rho}$$

$$\updownarrow$$

$$\boxed{\mathrm{W}^1[f] = 1}$$

Exercise

$$\updownarrow$$

$$\boxed{f = \pm\chi_i \text{ for some } i \in [n]}$$

a dictator or its negation

for $f : \{-1,1\}^n \to \{-1,1\}$,
$-1 < \rho < 0$

Beyond the impossibility result

Getting further insight via Fourier analysis

$$\lim_{n\to\infty} \mathrm{Stab}_\rho(\mathrm{Maj}_n) = 1 - \frac{2}{\pi}\arccos(\rho)$$

Using $f = \mathrm{Maj}_n$:

- $\lim_{n\to\infty} \Pr[\exists$ Condorcet winner$] = \frac{3}{2\pi}\arccos\left(-\frac{1}{3}\right) \approx 91.2\%$

Guilbaud's formula

Beyond the impossibility result

Getting further insight via Fourier analysis

$$\lim_{n\to\infty} \mathrm{Stab}_\rho(\mathrm{Maj}_n) = 1 - \frac{2}{\pi}\arccos(\rho)$$

The bound attained by the majority function is already nearly the best possible.

Using $f = \mathrm{Maj}_n$:

- $\lim\limits_{n\to\infty} \Pr[\exists$ Condorcet winner$] = \frac{3}{2\pi}\arccos\left(-\frac{1}{3}\right) \approx 91.2\%$

Using any $f : \{-1,1\}^n \to \{-1,1\}$ with all $\hat{f}(\{i\})$ equal:

- $\lim\limits_{n\to\infty} \Pr[\exists$ Condorcet winner$] \leq \frac{7}{9} + \frac{4}{9\pi} \approx 91.9\%$

An upper bound for any **transitive-symmetric** function.

Robustness of the impossibility result

$\Pr[\exists \text{ Condorcet winner}] \in 1 - \Theta(\epsilon)$

$\updownarrow$ Exercise

$\mathrm{W}^1[f] \in 1 - \Theta(\epsilon)$

$\updownarrow$ Friedgut–Kalai–Naor (FKN) theorem

f is $\Theta(\epsilon)$-close to $\pm\chi_i$ for some $i \in [n]$

Outlook

- **Next:** spectral structure + learning.

How to learn an unknown Boolean function via queries?

What structures enable efficient learning?

Boolean functions with sufficiently simple Fourier spectra can be efficiently learned.

References

- Sections 2.4–2.5 of https://arxiv.org/abs/2105.10386